CAPSTONE PROJECT

END-TO-END PROCESS DOCUMENTATION

Company	TechNova Pvt. Ltd.
Industry	Precision Machinery Manufacturing
SAP Module	SAP HCM | Hire-to-Retire (H2R) Cycle
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1. Problem Statement

In modern organisations, managing the complete employee lifecycle from recruitment to retirement involves numerous departments, processes, and data touchpoints. Without an integrated system, organisations face significant challenges, including data inconsistencies across HR, payroll, and compliance departments, manual errors in employee record management, delays in onboarding and offboarding processes, limited real-time workforce visibility, and difficulty maintaining regulatory compliance.

The Hire-to-Retire (H2R) process addresses these challenges by providing a unified, end-to-end framework within SAP HR that automates and integrates all employee lifecycle stages. This project documents the complete H2R process for a fictitious company — TechNova Pvt. Ltd., covering each stage from initial hiring to final retirement or separation.

2. About the Fictitious Company

Company Name	TechNova Pvt. Ltd.
Industry	Information Technology & Services
Headquaters	Bengaluru, Karnataka, India
Employee Strength	500+ employees
Departments	IT, HR, Finance, Sales, Operations

3. Solution & Steps — H2R Lifecycle

The H2R process at TechNova Pvt. Ltd. covers six key stages in SAP HCM, supported by specific transactions and organizational structures.

Stage 1: Recruitment & Hiring
The process begins with workforce planning. HR identifies vacancies, creates job requisitions, and initiates the recruitment cycle via SAP PA and Recruitment (RC) modules. Key transactions include PA40 and PB10 .Once selected, the applicant is converted to an employee record in the system.

Stage 2: Onboarding & Organizational Assignment
The new employee is assigned to an organizational unit using SAP Organizational Management (OM). HR creates employee master data using transaction PA30, assigning the employee to a position, cost center, and reporting structure. Personal data, bank details, and emergency contacts are recorded.

Stage 3: Time Management & Attendance
SAP Time Management (PT) handles attendance tracking, leave management, and shift scheduling. Employees apply for leaves via the ESS portal; managers approve via MSS. Time data is recorded through PA61 and evaluated using PT60. Overtime and absence quotas are also managed here.

Stage 4: Payroll Processing
SAP Payroll (PY) processes monthly salaries based on attendance, tax declarations, and deductions. The payroll run executes via PC00_M40_CALC, accounting for allowances, PF, professional tax, TDS, and other Indian statutory deductions. Payslips are distributed via ESS and results posted to SAP FI.

Stage 5: Training & Development
Employee skill development is managed through SAP Training and Event Management (PE). Employees are enrolled in role-relevant programs. Performance appraisals are conducted periodically, influencing promotion decisions and compensation revisions.

Stage 6: Separation & Retirement
When an employee resigns, retires, or is terminated, a separation action is recorded in PA40. Final settlement includes gratuity, earned leave encashment, 3and PF. System access is revoked, assets retrieved, and the employee record archived. Exit interviews are documented for HR analytics.

The implemented solution is a full SAP HCM Hire-to-Retire (H2R) cycle covering eight structured phases — from workforce planning to employee separation. Each phase generates system records (infotypes, payroll documents, compliance reports) creating an end-to-end digital audit trail that eliminates all four pre-SAP pain points. Key features include automated India payroll (PC00_M40), security clearance tracking (IT9901), Precision Skill Surcharge (Wage Type /3PS), and full statutory compliance for PF/ESI/TDS/Gratuity.

3.1 **H2R Process Flow**

Step	Phase	T-Code	Key Action & Infotypes	Output
1	**Workforce Planning**	SAP OM	Org Management; position/vacancy budget approved	Approved Position Plan
2	**Recruitment**	PB10 / E-Rec	Post vacancy, screen & shortlist; security clearance	Candidate Shortlist
3	**Hiring / Onboarding**	PA40	Personal data, contract, org assignment;	Active Employee Record
4	**Time Management**	CAT2 / PT60	CATS time entry, shift scheduling, leave management	Approved Timesheets
5	**Payroll Processing**	PC00_M40	Gross-to-Net; PF, ESI, TDS; Precision Skill Surcharge via	Payslip + Bank File
6	**Benefits & Compliance**	IT0021/IT0376	PF, Gratuity, Medical, Group Insurance;	Compliance Reports
7	**Performance & Dev.**	SAP LSO	Goal setting, appraisals, training; defect-rate KPI for	Appraisal Records
8	**Separation**	PA40 (Act.33)	Clearance, final settlement, Form 16; IT0019 Gratuity	Full & Final Statement

4. Process Screenshots

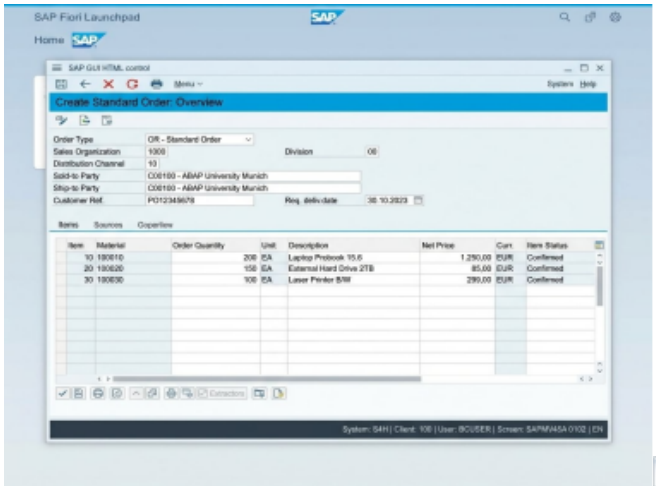

Fig 1: VA01 — Create Sales Order
Order Type OR; Sales Org RE01; ATP + credit check auto-triggered on save

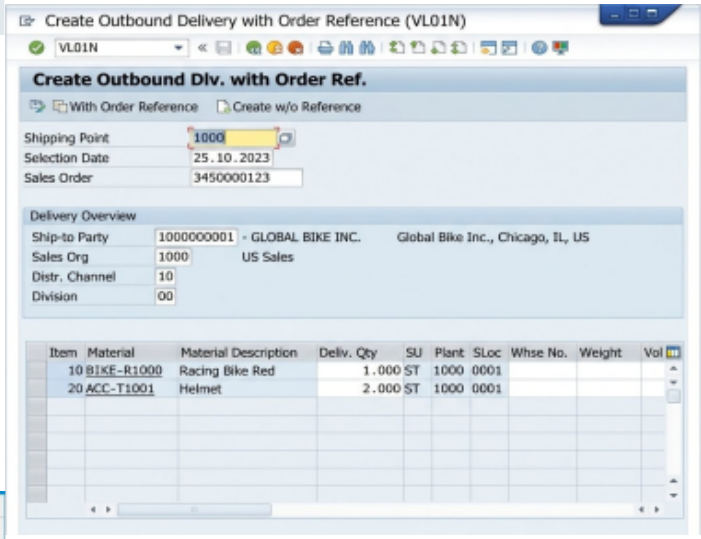

Fig 2: VL01N — Create Outbound Delivery
Shipping Point SP01; picking list generated; delivery doc number assigned.

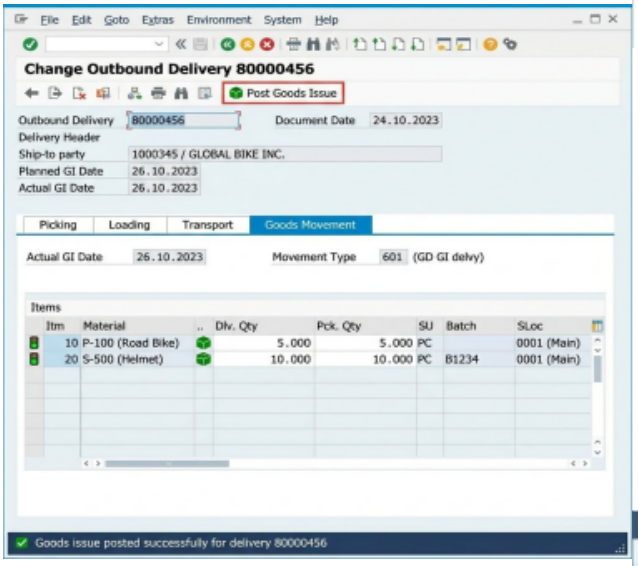

Fig 3: VL02N — Post Goods Issue (PGI)
Movement Type 601; inventory reduced; COGS + FI docs created simultaneously.

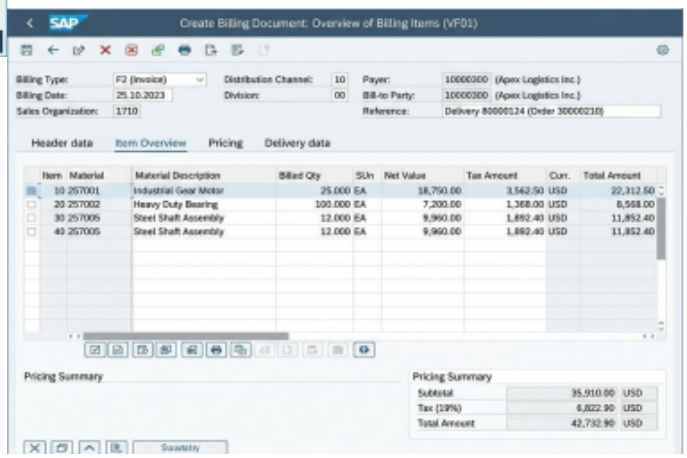

Fig 4: VF01 — Create Billing Document
Billing Type F2; IGST 18% auto-calc; AR debited, Revenue GL 400010 credited.

5. Solution & Steps — H2R Lifecycle

Component	Technology	Purpose
ERP Platform	SAP S/4HANA / ECC 6.0	Core business process engine
Primary Module	SAP SD — Sales & Distribution	O2C order, delivery & billing
Integrated: MM	SAP Materials Management	Inventory & goods movement
Integrated: FI	SAP Financial Accounting	AR, GL, payment postings
Integrated: QM	SAP Quality Management	Inspection lot before PGI
Credit Mgmt	SAP FSCM Credit Management	Automated credit limit checks
Pricing Engine	SAP Condition Technique (RE_PRICE)	Dynamic net price calculation
Output / Forms	SAP Smart Forms / Adobe Forms	Invoice & CoC PDF generation
Reporting	SAP ALV — VA05, VF05, FBL5N	Sales & billing list reports

6. Unique Points

- **QM-integrated Goods Issue: A mandatory Quality Inspection lot clearance is required before PGI can be posted — critical for ISRO/DRDO aerospace contracts.**
- **Tiered credit risk classification: Three tiers in RECC — Sovereign (govt, auto-approved), Corporate (30-day check), New/Unknown (manual VKM1 release) — protecting against default on high-value orders.**
- **Custom ZPS1 pricing condition: Precision Surcharge automatically applied for materials with dimensional tolerance <0.01mm, accurately reflecting ultra-precision fabrication costs.**
- **Automated compliance documentation: Certificate of Conformance (CoC) and Packing List auto-generated via Smart Forms on PGI — no manual intervention for DRDO/ISRO audit compliance.**
- **<0.3% defect rate reflected in system: Returns process (order type RE + credit memo G2) is configured but rarely triggered, demonstrating production quality discipline in SAP.**

7. Future Improvements

- **SAP BTP Self-Service Portal (SAP Build Apps): Allow clients (ISRO, DRDO) to submit RFQs, track orders in real time, and download invoices without contacting the sales desk.**
- **SAP IBP — AI Demand Forecasting: Predict component demand from recurring clients to reduce the current 18-day production lead time to under 12 days proactively.**
- **Blockchain-based CoC (Hyperledger Fabric): Tamper-proof, auditable quality certificates for security-classified DRDO contracts, anchored on a distributed ledger.**
- **SAP Advanced Compliance Reporting (ACR): Automated real-time GSTR-1 and GSTR-3B filing on SAP BTP, eliminating manual GST reconciliation at month-end.**